MTH209: Midsem Exam - Question 3

1. Follow the instructions **exactly**.
2. If your code is not properly commented or horizontal/vertical spacing is missing, points will be deducted.
3. The only libraries you are allowed to use are `uwot`, `Rtsne`, and `imager`.

Question

The goal of this problem is to device some common-sense method to predict the number written in an image. Please read the below carefully:

A dataset on the hand drawn images of numbers are provided in the Rdata file. Run the following code to load the Rdata file that contains two matrices: `digits_train` and `digits_test`:

```
load("digits.RData")
dim(digits_train)
```

```
[1] 2000  257
```

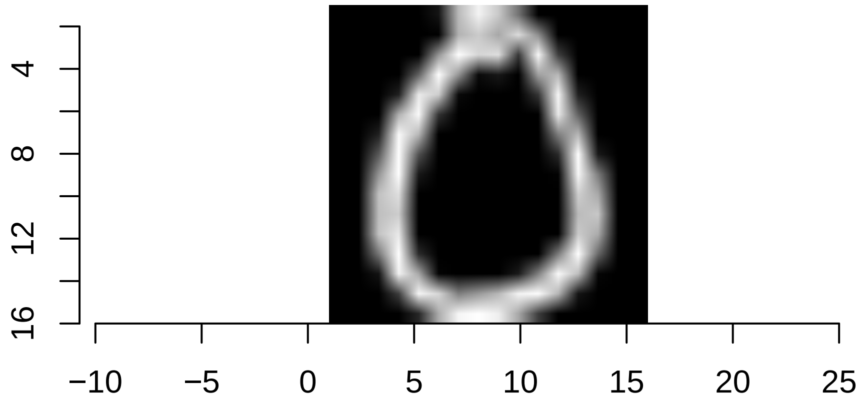
```
dim(digits_test)
```

```
[1] 1000  256
```

Note that the `digits_train` contains 2000 images where each row is an image and is 256 + 1 dimensional, since the first column of the dataset is the number in the image and the next 256 columns are the 16x16 images stacked in a row. However, `digits_test` contains 1000 images without their true label, and only the 16x16 = 256 pixel values are available.

Let us understand how these images are stored by recreating them:

```
library(imager)
# remove the first column
foo <- matrix(digits_train[1, -1], nrow = 16, ncol = 16, byrow = FALSE)
plot(as.cimg(foo))
```




```
# reveal the first column
digits_train[1, 1]
```

```
[1] 0
```

Your goal is to write code that *predicts* the label for the 500 images. Here's one way to do this (you may use any method):

1. Apply some appropriate dimension reduction to the combined 3000 numerical observations.
2. Obtain the embedding space values for each observation.
3. Device a strategy to *predict* the label for unknown observations in `digits_test` using the embedding mapping space.

You may use any other appropriate strategy as well. The goal is finally to obtain a prediction label vector of length 1000 called `prediction` for each image in `digits_test`. The code should look like the following

```
load("digits.RData")

# please only include what is necessary to reduce run time
...
# name must be prediction
prediction <- numeric(length = 1000)
...
```

I will be using your vector prediction to calculate the accuracy of your prediction and marks will be given based on accuracy.

Submission Instructions

INSTRUCTIONS: copy and paste all the code needed to obtain your `prediction` vector in a file called `question3.R` and upload it to mooKIT. This question has 30 marks.